No detectable edge: cointegration pairs trading on the Russell 3000, 2023–2025

The 300 most liquid stocks of an approximated Russell 3000, traded as cointegrated pairs with next-day fills and a per-stock cost model, July 2023 to December 2025.

Navnoor Bawa · Note of 20 October 2026 · Study run v32.0.0 of 18 October 2026, commit 711b18c

Code and data github.com/NavnoorBawa/russell3000-pairs-trading
Research-note site navnoorbawa.github.io/russell3000-pairs-trading

Summary

NO DETECTABLE EDGE

No detectable edge. Net of modelled costs the strategy's Sharpe ratio was −0.07 in the main backtest (95% interval −1.39 to +1.23) and −0.33 in a continuous walk-forward (−1.49 to +0.86), which is out of sample for pair selection and the ranker but not for the trading thresholds. Neither is distinguishable from zero, from the Gatev distance method (difference −0.14, p = 0.879) or from random pairs traded with that method (28th percentile). Before costs the trades earned +8.9 bp of the notional they traded (t = 0.28), less than the 12.9 bp of modelled cost: there is no detectable gross edge for costs to remove. The sample excludes Sharpe ratios of 1.5 and above; at 1.0 it is borderline (p = 0.046 assuming independent returns, 0.057 allowing for their autocorrelation), and it cannot tell 0 from 0.5. The decisive test is a 2026 holdout, to be pre-registered before any 2026 price series is fetched.

MAIN BACKTEST · NET SHARPE

−0.07

95% interval −1.39 to +1.23 · Newey–West p = 0.917 · 628 days

WALK-FORWARD · NET SHARPE

−0.33

95% interval −1.49 to +0.86 · Newey–West p = 0.559 · 711 days

UPPER BOUND ON THE SHARPE RATIO

1.04

One-sided 95%, allowing for autocorrelation (0.97 if independent) · 80% power only at 1.68

GROSS EDGE PER TRADE

+8.9 bp

Before costs, notional-weighted · t = 0.28 · modelled cost 12.9 bp · n = 161

BREAK-EVEN COST MULTIPLE

0.69×

Net P&L is zero at this fraction of modelled costs: +8.9 over 12.9 bp

DEFLATED SHARPE · WALK-FORWARD

0.4%

Probability that the true Sharpe ratio exceeds the expected best of 32 null trials · P(SR > 0) = 28.7%

1 Introduction

1.1 The question

Pairs trading buys one stock and sells another when the spread between them is unusually wide, and closes the position when it narrows. Gatev et al. (2006) documented excess returns from a simple distance rule on US stocks over four decades; later studies find that the edge has shrunk and is small after realistic costs (Do and Faff 2010; 2012; Rad et al. 2016), and Krauss

(2017) surveys the methods. This note asks one question and is built to answer it honestly: **does a cointegration pairs-trading strategy on the Russell 3000 have an edge after realistic execution and costs, out of sample?**

The strategy selects pairs with the Engle–Granger test (Engle and Granger 1987) on formation data only, tracks each pair's hedge ratio with a two-state Kalman filter (Kalman 1960), trades a z-score of the hedged spread with fills at the next day's close, and pays costs computed from each stock's own liquidity. It is evaluated in a main backtest from 3 July 2023 to 31 December 2025, in a continuous walk-forward with 11 test windows, and against two baselines run on the same footing: the distance method of Gatev et al. (2006) and 200 portfolios of random pairs traded with that method.

The answer for 2023–2025 is a null. A null is informative only to the extent that the test had power and the measurement was right, so most of this note is about what the sample can and cannot rule out, and about the checks that the code measures what it claims. The development period cannot test the design choices made on it; a 2026 holdout, sealed at the date of this note, does that (Section 10).

1.2 What is new

The study uses free data only. Academic studies avoid survivorship with paid point-in-time databases; a backtest on free data usually starts from today's index members instead, and is exposed to the other familiar weaknesses of backtests: flat cost assumptions, rolling windows that drop positions at their edges, significance without power, and an untold number of designs tried. Each item below addresses one of these; the last adds a test the development data cannot provide.

- **A point-in-time universe from free SEC filings.** The complete holdings of the iShares Russell 3000 ETF (IWV) are public on Form N-PORT for the last month of each fiscal quarter; 26 quarterly snapshots give index membership as it was at each formation date, without a paid constituent history (Section 2). The v33 design trades this universe; the v32 run reported here does not.
- **Costs from each stock's own liquidity.** Every fill pays commission, half the typical quoted spread for the stock's dollar-volume bucket and square-root market impact on the stock's own volatility and dollar volume, at the next day's close (Section 7).
- **A continuous walk-forward.** The test windows are calendar slices of one simulation with the ranker refit per window; positions carry across window boundaries instead of being discarded (Section 3).
- **Power analysis.** Each null is reported with the Sharpe ratios it excludes, with and without allowance for autocorrelation, the Sharpe ratio it could have detected and the track record a given Sharpe ratio would need (Section 5).
- **A declared variant family.** On 20 October 2026, before any variant ran, the study declared 27 threshold settings and 11 design toggles for White's Reality Check, Hansen's SPA test, the Romano–Wolf stepdown, the model confidence set and the probability of backtest overfitting (Section 5.3).

– **A 2026 holdout.** Hypotheses, test, multiplicity correction and decision rule are to be fixed in a pre-registration before any 2026 price series is fetched; each strategy then runs once and its result is published whatever it is (Section 10).

1.3 Related work

The published record points one way: the classic pairs edge was large before the 2000s, has shrunk since, and is small after realistic costs (Table 1). The closest analogue of this study is Rad et al. (2016), who compare distance, cointegration and copula methods on US stocks; this study differs in period, cost model and inference. Avellaneda and Lee (2010) trade residuals of single stocks against sector ETFs and factors, the second strategy family of this study (Section 9).

TABLE 1
Published results for pairs trading and residual reversion, as reported by their authors

Study	Sample	Method	Reported result
Gatev et al. (2006)	US, 1962–2002	Distance method, top-20 pairs	Annualised excess returns of up to 11% before costs, falling in later years. The baseline replicated here.
Do and Faff (2010)	US, to 2009	Distance method	Profitability declines over time; the strategy does best in prolonged turbulence.
Avellaneda and Lee (2010)	US, 1997–2007	PCA and sector-ETF residual reversion	Net Sharpe 1.44 (PCA) and 1.1 (ETF) over 1997–2007; PCA only 0.9 over 2003–2007. The second family tested here.
Do and Faff (2012)	US, 1963–2009	Distance method with trading costs	About 30 bp a month for well-matched pairs in refined industries after costs; largely unprofitable after 2002.
Rad et al. (2016)	US, 1962–2014	Distance, cointegration and copula methods	91, 85 and 43 bp a month before costs; 38, 33 and 5 bp after. From 2009 opportunities for the distance and cointegration methods fall, while the copula method's stay stable. The closest published analogue of this study.
Krauss (2017)	Review	Survey of pairs-trading approaches	Distance, cointegration, time-series, stochastic-control and other approaches compared; open problems include costs and robustness.

Source The studies cited.

2 Data

Every input is free: no account, API key or licence is needed to rebuild the study (Table 2). Prices are not redistributed with the code, under the vendor's terms; each run's fetch manifest lists the symbols and dates a re-run fetches.

TABLE 2
Data sources, all free

Source	What it provides	Used for
Yahoo Finance chart API	Daily open, high, low, close and volume, splits and dividends	Prices of the universe, the 11 sector SPDR ETFs and the VIX
SEC EDGAR: Form N-PORT filings of IWV	Quarterly holdings of the iShares Russell 3000 ETF	Point-in-time index membership (v33)
OpenFIGI mapping API	CUSIP to ticker	Joining holdings to price histories
SEC EDGAR: company submissions	SIC industry code per registrant	Fama–French 49 industries for same-industry pairs (v33)
Kenneth French data library	Fama–French five factors, momentum, short- and long-term reversal, one-month T-bill	Factor attribution (v33)
Nasdaq stock screener	Market capitalisation of US-listed common stocks	The v32 universe

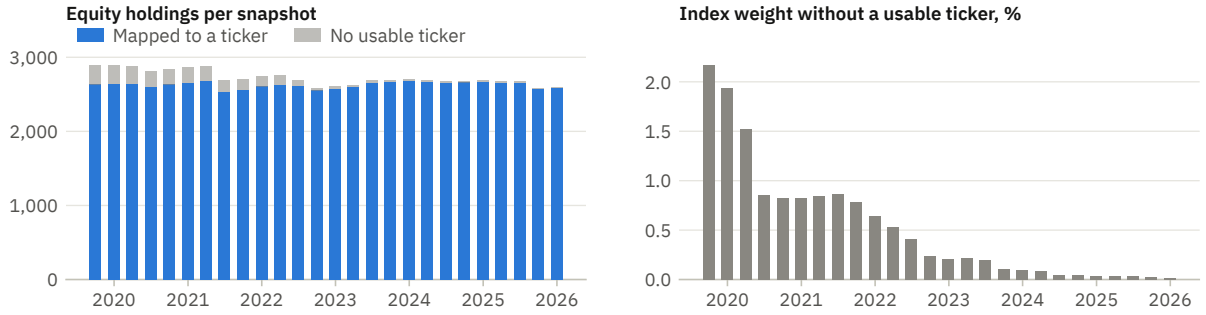
2.1 The v32 universe

The universe of the v32 study run is the 3,000 largest US common stocks by market capitalisation on 18 October 2026: the Russell 3000 rule applied to the Nasdaq screener, not FTSE Russell's constituent list. Dropping invalid and duplicate rows leaves 2,999 symbols. Daily prices from 2 January 2020

EXHIBIT 1

Free filings give 26 quarterly snapshots of index membership; the index weight without a usable ticker falls from 2.2% to 0.0%

Equity holdings of the iShares Russell 3000 ETF in its N-PORT filings, 2019–2025, and the share of their value that cannot be mapped to a ticker



Notes A holding without a usable ticker is one OpenFIGI cannot map, or one whose last ticker now belongs to another issuer and is left unmapped on purpose so that no member is joined to someone else's prices. The share is larger in older snapshots, as expected if most such holdings are securities that have since delisted; its weight is part of the survivorship gap.
Source SEC EDGAR, Form N-PORT-P filings of the iShares Russell 3000 ETF (iShares Trust, series S000004341); tickers from OpenFIGI; author's calculations.

to 31 December 2025 were obtained for 2,752 of them (91.8%). Of the rest, 86 fetches failed with HTTP errors, 86 histories failed the data checks (missing or non-positive closes, or a point-in-time price below $2 when the series starts) and 75 were too short; the failed fetches are a shortfall of the run, not a limit of the data. The level screens, a $5 price floor and a dollar-volume floor, are applied to point-in-time prices: closes with later splits undone, so that a split after the formation date cannot decide earlier membership. The 300 most liquid names by point-in-time dollar volume enter the pair search.

This universe has survivorship bias. Companies that failed, delisted or were acquired before 18 October 2026 are absent, and so are the pairs that broke when one of them failed. The bias flatters a mean-reversion strategy, which makes a null result on this universe more convincing, not less (Section 11). The ranking by market capitalisation on that date is also one of two pieces of 2026 market information in the study (Section 10).

2.2 Point-in-time membership from N-PORT filings

A registered fund reports its complete portfolio holdings on Form N-PORT, and the report for the third month of each fiscal quarter is made public on EDGAR. IWV tracks the Russell 3000, so its filings give index membership as it was, quarter by quarter, at no cost. The study reads the equity-common holdings of 26 snapshots from 30 September 2019 to 31 December 2025, 2,581 to 2,895 holdings each, and maps each holding's CUSIP to a ticker through OpenFIGI (Exhibit 1). A snapshot is in force from the date its holdings describe until the next one. Two caveats: IWV is a fund, not the index, and holds fewer names than the index; and additions and deletions within a quarter appear only at the next quarter end. Snapshots dated 1 January 2026 or later are listed but not downloaded: their holdings carry a market value, and so a 2026 price, for every member.

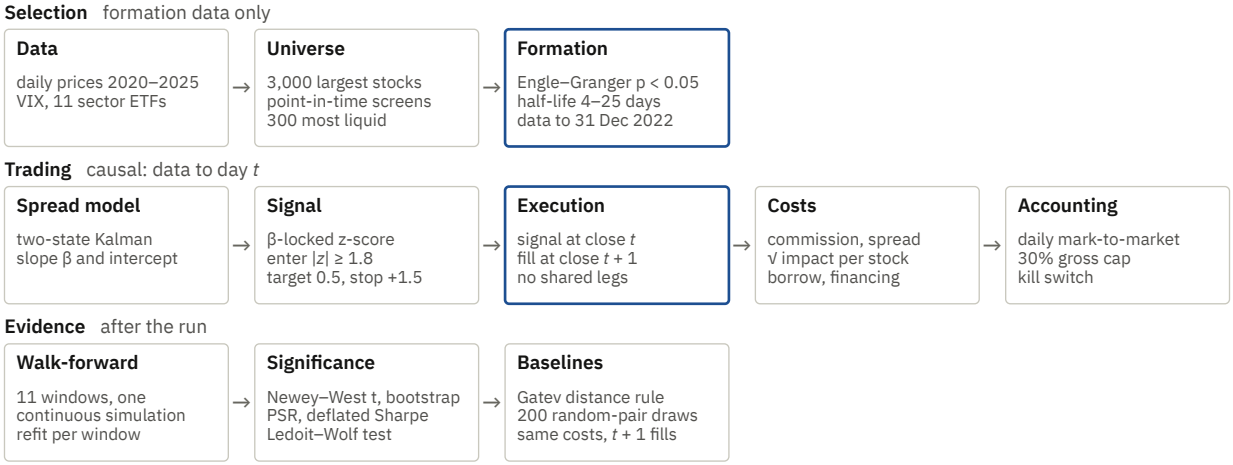
2.3 The survivorship gap

The survivorship gap at a formation date is the index weight of members a backtest cannot trade: holdings without a usable ticker, plus members with

EXHIBIT 2

The pipeline, and what data each stage may see

Stages of one run, grouped by the data each may see

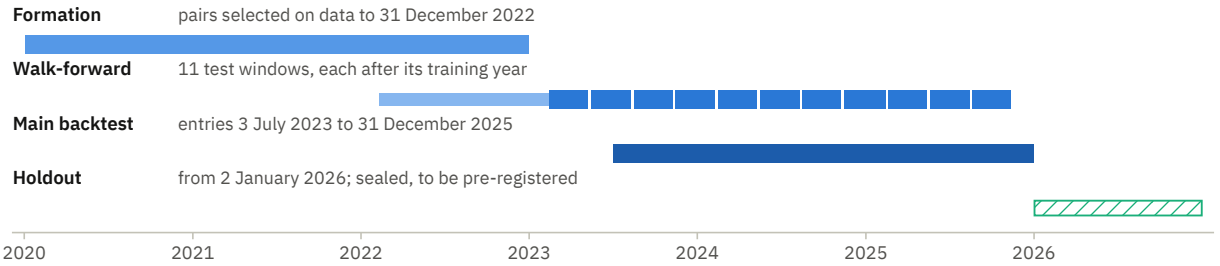


Notes Boxes framed in the accent colour mark the two look-ahead boundaries: the formation cutoff, after which selection sees no data, and the next-day fill, which separates a signal from its trade.
Source Author's design of the v32.0.0 study run.

EXHIBIT 3

Data splits: formation, walk-forward, main backtest and the sealed holdout

Calendar 2020–2026



Notes The main backtest and the walk-forward overlap on 616 of the main backtest's 628 days and trade the same pair pool: they are one experiment, not two confirmations.
Source Author's study calendar for the v32.0.0 run.

a ticker whose price history the vendor no longer serves, mostly failures and takeovers. The v32 universe has no membership to measure the gap against. From v33 on, every formation records the snapshot in force, its members, how many have price history, and the index weight of those that do not (forthcoming, Table 11).

3 Method

Selection sees formation data only; every trading decision on day t uses data up to day t and fills on day $t + 1$; the evidence is computed after the run (Exhibit 2). Tests change future prices and check that no past decision moves. Table 12 in the appendix states every rule of the run on one page, and Table 13 sets the v32 design beside the v33 design.

3.1 Formation

Pairs are formed once, on data to 31 December 2022. Among the 300 most liquid names, every pair with a positive return correlation of at least 0.25 is tested for cointegration with the two-step Engle–Granger test in both directions, Bonferroni-adjusted across the two, at $p < 0.05$. A pair is kept if the

half-life of its cointegrating residual lies between 4 and 25 trading days and a heuristic quality score passes: 1,776 of 35,491 pairs tested (Section 5.4). Every 90 trading days the pool is re-tested on the trailing 2 years, and a pair stays only if it is still cointegrated there and its Engle–Granger p-value on the last 126 calendar days (about 87 trading days) is at most 0.30. Pairs can drop out but none can enter.

3.2 Spread model

The hedge ratio is the slope state of a two-state Kalman filter on log prices,

$$\log p_{1,t} = \alpha_t + \beta_t \log p_{2,t} + \varepsilon_t,$$

where (α_t, β_t) follows a random walk, both log prices are centred on their burn-in means and the filter is seeded by ordinary least squares. The intercept makes β a hedge ratio rather than a ratio of price levels, and centring makes every signal invariant to a vendor's back-adjustment of history for a later split or dividend, which shifts each log price by a constant.

3.3 Signal and exits

One z-score drives every decision. At a signal on day t_0 the hedge ratio is locked at its current value and the spread is standardised by its mean and standard deviation over the last 30 to 60 bars (three half-lives), both frozen at the signal:

$$s_t = \log p_{1,t} - \beta_{t_0} \log p_{2,t}, \quad z_t = \frac{s_t - \mu_{t_0}}{\sigma_{t_0}}.$$

A pair is entered when $|z| \geq 1.8$, long the spread when z is negative and short when it is positive. It exits at the target once z has reverted past −0.5 for a long spread or +0.5 for a short one, at the stop when z has moved 1.5 further against the position, and at the time stop after clamp(2.5 × half-life, 10, 25) trading days. Entries also require a 20-day return correlation of at least 0.20 and a hedge ratio in [0.25, 4]; after three consecutive losing trades a pair is barred from new entries for 30 calendar days; and a regime gate on the VIX and sector dispersion scales positions and blocks new entries in stressed markets.

3.4 Execution, accounting and risk limits

A signal computed on the close of day t is filled at the close of day $t + 1$, for entries and exits alike: in practice a market-on-close order in each stock's closing auction, so both legs print together. Share counts are fixed from the signal-day close and the legs are sized \$1 : \$β. Positions are marked to market daily, costs are booked on fill days, and positions still open at the end of the data are closed. The book has \$100M of notional capital, 3–10% of equity per pair before regime scaling (1.2% to 9.0% as traded), a 30% cap on gross exposure, at most 8 new entries a day, no stock in two open pairs, and drawdown and daily-loss kill switches that flatten it. Idle cash earns no interest.

A classifier of the probability that a trade is profitable, on 30 features, only orders same-day candidates, which matters when more signals fire than the book's limits allow; it does not create signals. It has no economically useful

skill: its out-of-sample AUC is 0.514 (0.510 pooled across the walk-forward windows, where 0.5 is no skill), and its pooled Brier score, 0.2547, is worse than the 0.2499 of always predicting the base rate. The v33 design drops it.

3.5 Walk-forward

The walk-forward is one continuous simulation on a 252-day training / 63-day test grid from February 2023 to December 2025; the last window runs to the walk-forward's last day, 12 December 2025, 81 trading days. It is out of sample for pair selection, made on data to 31 December 2022, and for the ranker, refit on each window's training year. It is not out of sample for the entry, exit and stop thresholds, which were chosen across versions on this same data, or for the regime gate, whose thresholds were fitted on windows that now lie inside this period. This note therefore calls it the walk-forward and reserves "out of sample" for the 2026 holdout. Positions carry across window boundaries, and per-window figures are calendar slices of the one simulation (Table 15 in the appendix). It trades the same pair pool as the main backtest and shares 616 of its 628 days (Exhibit 3).

3.6 Inference

Sharpe ratios are annualised from daily net returns by $\sqrt{252}$. Means are tested with the t-statistic of Newey and West (1987). Intervals for single series are circular block bootstraps with 5,000 resamples and the block length of Politis and White (2004), as corrected by Patton et al. (2009): blocks of 3 days for the main backtest and 2 for the walk-forward. The probabilistic and deflated Sharpe ratios follow Bailey and López de Prado (2012, 2014), using the variance of the Sharpe estimator under non-normal returns. Standard errors, power and the upper bound use that variance twice: assuming independent returns, and allowing for the returns' lag-1 autocorrelation (Section 5). The autocorrelation-adjusted annualisation of Lo (2002), which the note does not otherwise use, gives −0.07 for the main backtest and −0.36 for the walk-forward. Differences of Sharpe ratios use the studentized bootstrap of Ledoit and Wolf (2008).

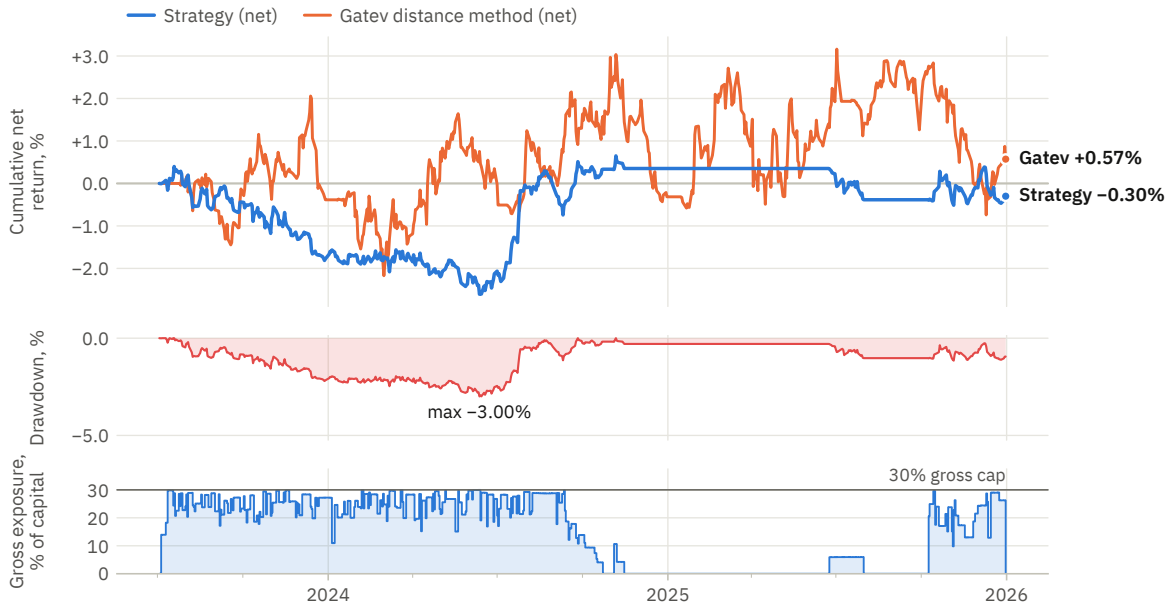
3.7 Validation

A backtest on real data says what a strategy did, not whether the code measures what it claims. A synthetic validation runs the real engine, with its full cost model, on markets whose answer is known: planted cointegrated pairs must show a large gross edge; correlated pairs that are not cointegrated, forced into the universe, must show none; the Kalman hedge ratio must recover the true slope; and pair selection must find the planted pairs while admitting at most 5% of unrelated ones. Its committed results are forthcoming (Table 11). The test suite adds causality checks, each confirmed by reintroducing the bug it guards against: every decision dated on or before T is unchanged when all prices, volumes and macro data after T are replaced, with classical ranking and with a trained ranker; the regime gate's inputs are prefix-invariant; and the same trades occur at the same z-scores when the vendor back-adjusts history for a later split. The statistics are tested against reference implementations: statsmodels for Newey–West, arch for the Poli-

EXHIBIT 4

Net of costs the book ends slightly below where it started; the Gatev baseline ends at +0.57%

Cumulative net return, underwater drawdown and gross exposure; main backtest, daily, July 2023 – December 2025



Notes The two paths are on different capital bases: the Gatev return is per unit of committed per-leg notional, at 4.55% annualised volatility, and the strategy's per unit of $100M of capital, at 1.53%. Compare their Sharpe ratios (Exhibit 8), not the levels of these paths. Gross exposure is rebuilt from the trades and reproduces the run's average and maximum (29.99%).
Source Author's backtest, v32.0.0 study run (commit 711b18c); prices from Yahoo Finance; baseline: Gatev et al. (2006) distance rule with the same $t + 1$ fills and costs.

tis–White block length, an independent cross-validation for the probability of backtest overfitting, and the authors' code for Ledoit–Wolf.

4 Development-period results

The main backtest enters trades from 3 July 2023 to 31 December 2025 on $100M of notional capital, with pairs selected on data to 31 December 2022. Its 161 trades returned −0.30% with a maximum drawdown of −3.00% (Exhibit 4). The book held no position on 207 of 628 days and used 15.1% of capital on average (gross), against a 30% cap. Table 3 sets out the standard statistics of the main backtest, the walk-forward and the Gatev baseline.

The walk-forward points the same way (Exhibit 5). Of its 11 windows, 3 made money, and the stitched path ends at −1.75%; the mean window return is −0.158% with a 95% interval from −0.636% to +0.320%. Two features of the main backtest deserve care. One month carries the result (Exhibit 6), and every half-year window whose Sharpe ratio looks significant overlaps that month (Exhibit 7); reading either as a regime the strategy could exploit would be fitting a rule to the test period.

5 Inference

Is the result significant? No: every interval in Exhibit 8 contains zero, and Table 5 gives the tests. The more useful question is what the data can rule out, and the answer depends on how the standard error of the Sharpe ratio treats dependence. With 628 daily observations, assuming independent returns, the standard error of the annualised Sharpe ratio is 0.63 and the one-sided

TABLE 3

Performance of the main backtest, the walk-forward and the Gatev baseline, net of costs

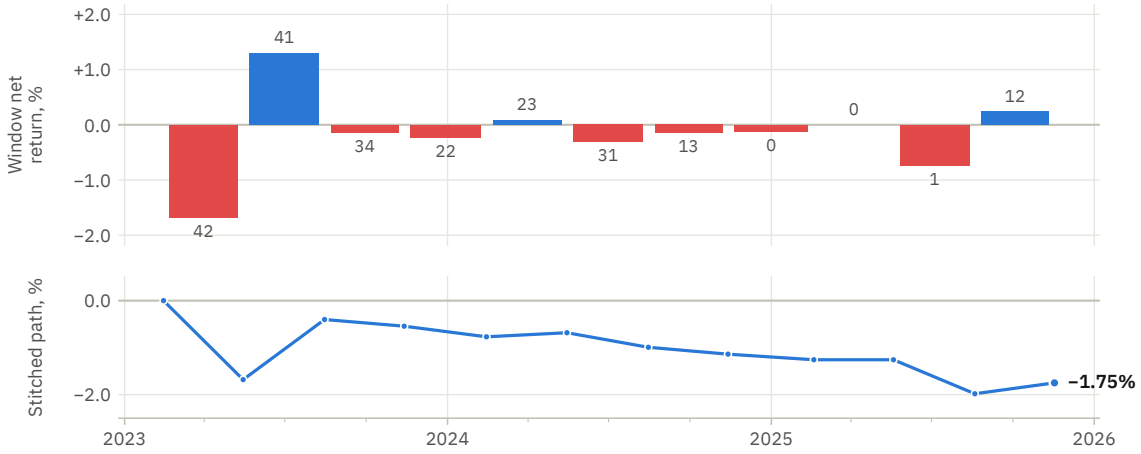
	Main backtest	Walk-forward	Gatev distance method
Trading days	628	711	628
Total net return, %	−0.30	−1.75	+0.57
Annualised net return, % (95% interval)	−0.11 [−2.15, +1.93]	−0.61 [−2.65, +1.43]	+0.33 [−4.87, +5.53]
Annualised volatility, % (95% interval)	1.53 [1.36, 1.70]	1.82	4.55 [4.11, 5.01]
Sharpe ratio (95% interval)	−0.07 [−1.39, +1.23]	−0.33 [−1.49, +0.86]	+0.07 [−1.15, +1.27]
Maximum drawdown, %	−3.00	−2.35	−4.14
Drawdown length, trading days: to trough / to recovery	227 / 300	—	53 / 191
Average gross exposure, % of capital	15.1	—	—
Net return on average gross exposure, % a year	−0.72	—	—
Trades	161	219	142
Hit rate, % of trades	47.2	48.0	—
Average holding period, trading days	12.6	—	55.5
Annual turnover, × capital	6.0	—	—

Notes Annualised returns are mean daily returns × 252 with a Newey–West 95% interval; volatility intervals are circular block bootstraps with the Sharpe ratio's block length. The walk-forward is exported as summary statistics only: its volatility is implied by its mean return and Sharpe ratio, and it has no volatility interval or drawdown length. The Gatev return is per unit of committed per-leg notional, the strategy's per unit of capital; compare Sharpe ratios across columns, not return or volatility levels. A dash marks a statistic the run does not report for that series.
Source Author's backtest, v32.0.0 study run (commit 711b18c); prices from Yahoo Finance; baseline: Gatev et al. (2006) distance rule.

EXHIBIT 5

In the walk-forward, 3 of 11 windows made money; the stitched path ends at −1.75%

Net return per test window (bars, labelled with the trades opened) and the compounded path at window ends; February 2023 – December 2025



Notes Out of sample for pair selection and the ranker only (Section 3). The windows are calendar slices of one continuous simulation, 63 trading days each and 81 for the last; positions carry across window boundaries, so a window that opens no trade can still carry P&L from earlier positions (window 8). No trade was opened in 2 of the windows. With 11 windows the test has little power: mean −0.158% per window, 95% interval −0.636% to +0.320%, p = 0.478.
Source Author's backtest, v32.0.0 study run (commit 711b18c); prices from Yahoo Finance.

EXHIBIT 6

One month carries the result: July 2024 earned +2.14%; without it the period returned −2.39%

Monthly net return, % of capital; the year column compounds the months shown

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
2023							−0.08	−0.35	−0.01	−0.74	−0.12	−0.48	−1.78
2024	+0.10	−0.15	+0.11	−0.46	+0.04	−0.01	**+2.14**	−0.01	+0.41	−0.04	+0.02	0.00	**+2.17**
2025	0.00	0.00	0.00	0.00	0.00	−0.12	−0.61	0.00	0.00	+0.41	−0.12	−0.21	−0.65

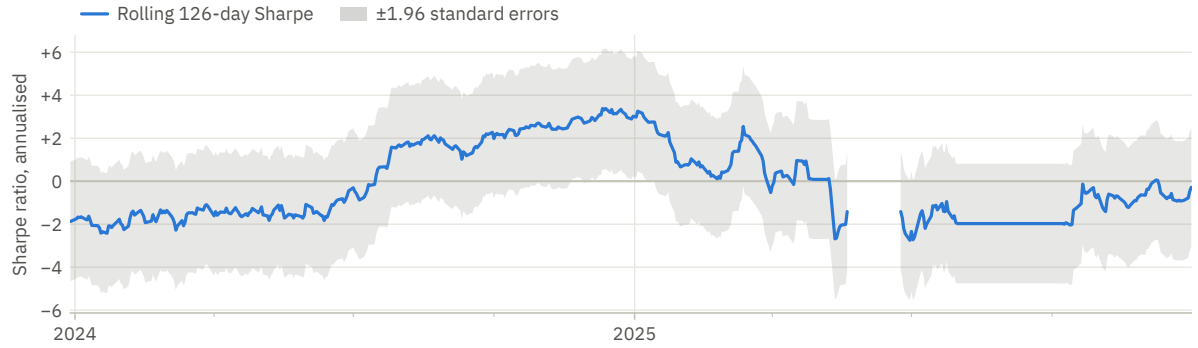
Notes Of the 30 months, 7 were positive; compounded, they return −0.30%. The book was flat from December 2024 to May 2025; the regime gate blocked new entries on 248 of the 628 days. This grid is the evaluated period: a rule fitted to it would be data snooping.
Source Author's backtest, v32.0.0 study run (commit 711b18c); prices from Yahoo Finance.

95% upper bound 0.97; allowing for the returns' lag-1 autocorrelation of 0.07 they are 0.68 and 1.04. Either way the sample excludes Sharpe ratios

EXHIBIT 7

Half-year Sharpe ratios swing between −2.8 and +3.4; every window whose band clears zero overlaps July 2024

Rolling 126-day Sharpe ratio, annualised, with a ±1.96 standard-error band; main backtest

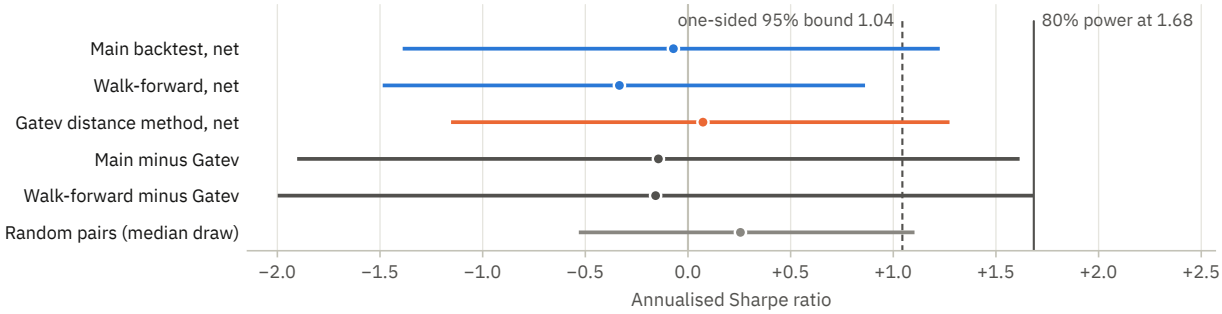


Notes SE = √((1 + SR²/2)/*n*) per day with *n* = 126, annualised, the independent-returns standard error of Lo (2002). Of the 503 overlapping windows, 30 clear zero, all ending between 21 November 2024 and 8 January 2025; overlapping windows are not independent tests. Windows in which the book was flat throughout have no Sharpe ratio.
Source Author's calculation from the daily net returns of the v32.0.0 run.

EXHIBIT 8

Every Sharpe interval spans zero; the data exclude only Sharpe ratios above about 1

Annualised Sharpe ratio with its 95% interval; development period, net of costs



Notes Intervals are circular block bootstraps (Politis–White block length, 5,000 resamples) for single series and Ledoit–Wolf studentized bootstraps for differences; the random-pair row shows the 5th–95th percentile spread of 200 draws, not a confidence interval. The bound and the 80%-power line allow for lag-1 autocorrelation; assuming independent returns they are 0.97 and 1.58. The walk-forward's comparison with Gatev uses the 616 days the two series share.
Source Author's backtest, v32.0.0 study run (commit 711b18c); prices from Yahoo Finance; baseline: Gatev et al. (2006) distance rule, with the run's own bootstrap.

of 1.5 and above. A Sharpe ratio of 1.0 is borderline: it is rejected at the 5% level only if returns are independent (p = 0.046, against 0.057 allowing for autocorrelation). A Sharpe ratio of 0.5 is not rejected (p = 0.184 and 0.200). A true Sharpe ratio would need to be about 1.58, or 1.68 allowing for autocorrelation, to be detected with 80% probability. An observed Sharpe ratio of 0.5 becomes significant only after about 10.8 years of daily data (12.4 allowing for autocorrelation), and a strategy whose true Sharpe ratio is 0.5 reaches that bar only about half the time; at 1.0 the lengths are 2.7 and 3.1 years.

5.1 Power and the one-sided upper bound

A test that cannot reject zero is evidence of no edge only to the extent that it could have detected one. The power figures use the asymptotic variance of the Sharpe estimator for non-normal returns, with the sample's skewness (0.17) and non-excess kurtosis (6.7) (Bailey and López de Prado 2012), first for independent returns and then with the returns' lag-1 autocorrelation entering through an AR(1) extension of the same variance (Table 4). At this

TABLE 4

What the development sample rules out, assuming independent returns and allowing for autocorrelation

True annual Sharpe ratio x	p-value of H_0: SR ≥ x, independent	p-value, lag-1 autocorrelation	Years for an observed SR = x to be significant, independent	Years, lag-1 autocorrelation
0.5	0.184	0.200	10.8	12.4
1.0	0.046	0.057	2.7	3.1
1.5	0.007	0.010	1.2	1.4
2.0	0.001	0.001	0.7	0.8

Notes p-values are one-sided, evaluated at the observed Sharpe ratio. The track-record length is the minimum track record length of Bailey and López de Prado (2012) at the one-sided 5% level: the years of daily data after which an observed Sharpe ratio of x is significant. A strategy whose true Sharpe ratio is x reaches it only about half the time; 80% power needs about 2.3 times as long. The autocorrelation columns use the asymptotic variance for non-normal returns with lag-1 autocorrelation ρ = 0.07, $(a - b\cdot\gamma_3\cdot SR + c\cdot(\gamma_4 - 1)/4\cdot SR^2)/T$, where a, b and c depend on ρ and equal 1 at ρ = 0 (Section 13 gives its source).
Source Author's calculation from the daily net returns of the v32.0.0 run.

TABLE 5

Significance tests, v32.0.0 study run of 18 October 2026

Test	Main backtest	Walk-forward (daily)
Annualised Sharpe ratio	−0.07	−0.33
Newey–West t (mean ≠ 0)	−0.10	−0.58
Newey–West p	0.917	0.559
Probabilistic Sharpe, P(SR > 0)	45.5%	28.7%
Deflated Sharpe (32 trials)	1.4%	0.4%
Block-bootstrap 95% interval	[−1.39, +1.23]	[−1.49, +0.86]
Gatev distance method, Sharpe [95% interval]	+0.07 [−1.15, +1.27]	−0.01 (on 616 shared days)
Strategy minus Gatev, Sharpe (Ledoit–Wolf p)	−0.14 (0.879)	−0.16 (0.859)

Notes Newey–West t and p test a zero mean of daily net returns. The deflated Sharpe ratio is the probability that the true Sharpe ratio exceeds the expected maximum Sharpe ratio of the stated number of independent null trials. The walk-forward's comparison with Gatev uses the 616 days both series cover, on which the walk-forward's own Sharpe ratio is −0.17; no interval is computed for the Gatev Sharpe ratio on those days.
Source Author's backtest, v32.0.0 study run (commit 711b18c); prices from Yahoo Finance; tests of Newey and West (1987), Bailey and López de Prado (2014) and Ledoit and Wolf (2008).

sample length the critical Sharpe ratio of a one-sided 5% test is 1.04 (1.11 with autocorrelation) and the Sharpe ratio detected with 80% probability 1.58 (1.68); the one-sided 95% upper bound, 0.97 (1.04), is the largest Sharpe ratio the data do not reject. The sample therefore excludes a strategy of Sharpe 1.5 or 2, is borderline at 1, and says nothing useful about 0.5.

5.2 Baselines

The Gatev distance method, run on the same universe and period with the same $t + 1$ fills and cost model, returned +0.57% (Sharpe +0.07; 95% interval −1.15 to +1.27). Its return is per unit of committed per-leg notional, at 4.55% annualised volatility against the strategy's 1.53% on capital, so the Sharpe ratios, not the returns, are comparable (Table 3). The difference in Sharpe ratios is −0.14 in the main backtest (Ledoit–Wolf p = 0.879) and −0.16 in the walk-forward, on the 616 days the two series share (p = 0.859). Against 200 portfolios of random pairs, traded with the Gatev rule under the same costs, the strategy sits at the 28th percentile (Exhibit 9).

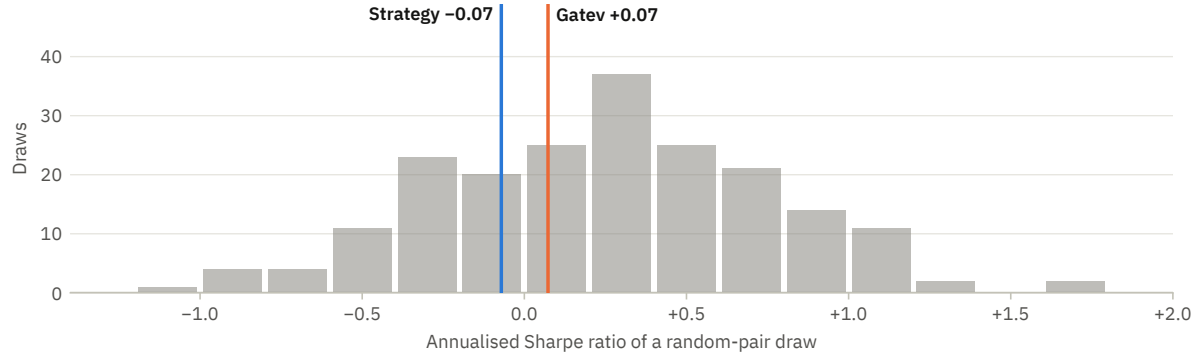
5.3 Deflated Sharpe and data snooping

Every design choice in this study was made on 2020–2025 data over 32 versions (Table 16). The deflated Sharpe ratio (Bailey and López de Prado 2014) is the probability that the true Sharpe ratio exceeds the expected maximum Sharpe ratio of N independent null trials; the run reports N = 32, a lower bound on the designs tried. At that N the probability is 1.4% for the main backtest and 0.4% for the walk-forward (Exhibit 10). Harvey et al. (2016) make the same point for the factor literature, arguing that a new factor's t-statistic should clear about 3 rather than 2.

EXHIBIT 9

The strategy sits at the 28th percentile of 200 random-pair portfolios

Annualised Sharpe ratio of random 20-pair draws traded with the distance rule under the same costs; main period

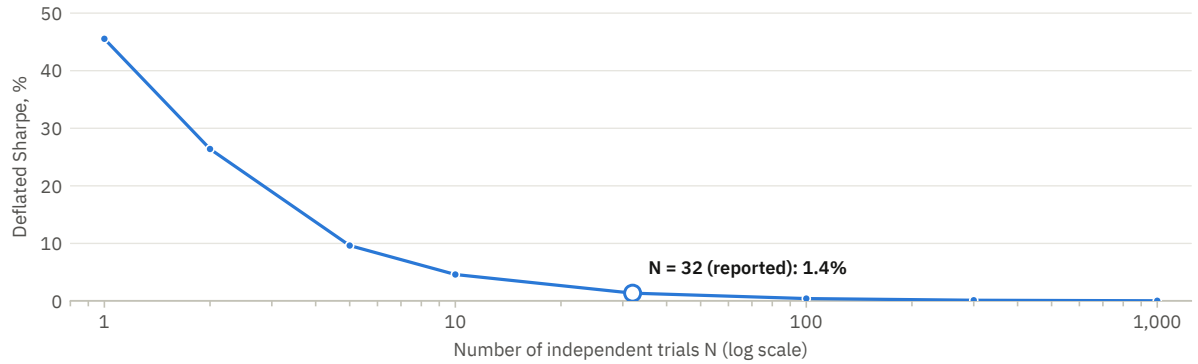


Notes Median draw +0.26, 5th–95th percentile −0.53 to +1.10. The control trades random pairs with the Gatev distance rule, not this strategy's engine, so it is a baseline for the distance method and only a rough one for this strategy.
Source Author's backtest, v32.0.0 study run (commit 711b18c); prices from Yahoo Finance.

EXHIBIT 10

Deflated for 32 trials, the main backtest's Sharpe ratio has a 1.4% probability of exceeding the expected best of the null trials

Deflated Sharpe ratio of the main backtest against the number of independent trials *N* (log scale)



Notes *N* = 1 is the probabilistic Sharpe ratio; the run reports *N* = 32, a lower bound on the designs tried over 32 versions. The trial Sharpe variance is the null variance of the estimator at this sample length.
Source Author's calculation from the daily net returns of the v32.0.0 run.

N is not known, and the deflated Sharpe ratio is only as good as the guess. The data-snooping tests replace the guess with a declared family. On 20 October 2026, before any variant ran, the study declared a 27-point grid of the entry, exit and stop thresholds and 11 design toggles around the v33 design (Table 14 in the appendix); each variant runs once on the development calendar and the primary design is not re-chosen from the results. The Reality Check of White (2000) and the SPA test of Hansen (2005) ask whether the best variant beats cash; the stepdown of Romano and Wolf (2005) identifies which variants do, with familywise error control; the model confidence set of Hansen et al. (2011) keeps the variants that cannot be told apart from the best; and the probability of backtest overfitting (Bailey et al. 2017) measures how often the in-sample winner falls below the median out of sample. Resampling uses the stationary bootstrap (Politis and Romano 1994). The results are forthcoming (Table 11).

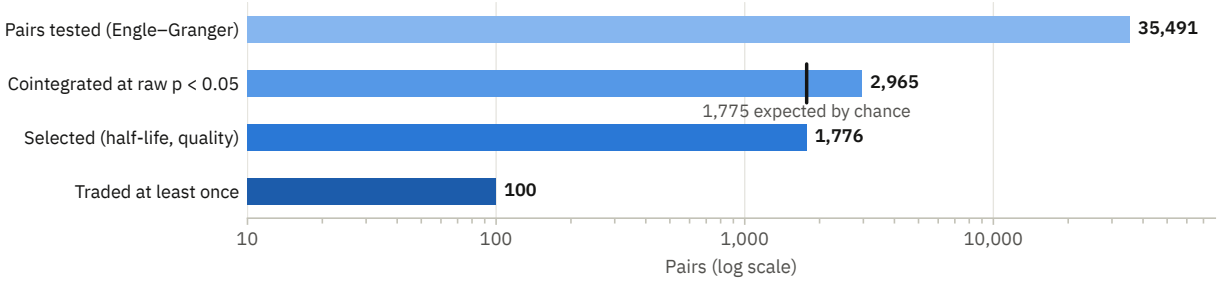
5.4 Multiple testing in pair selection

Pair selection is itself a multiple test. Of 35,491 pairs tested, 2,965 were "cointegrated" at $p < 0.05$. If no pair were cointegrated, about 1,775 would

EXHIBIT 11

From 35,491 candidate pairs to 100 traded; chance could account for most "cointegrated" pairs

Pairs at each selection stage, formation data to 31 December 2022; log scale

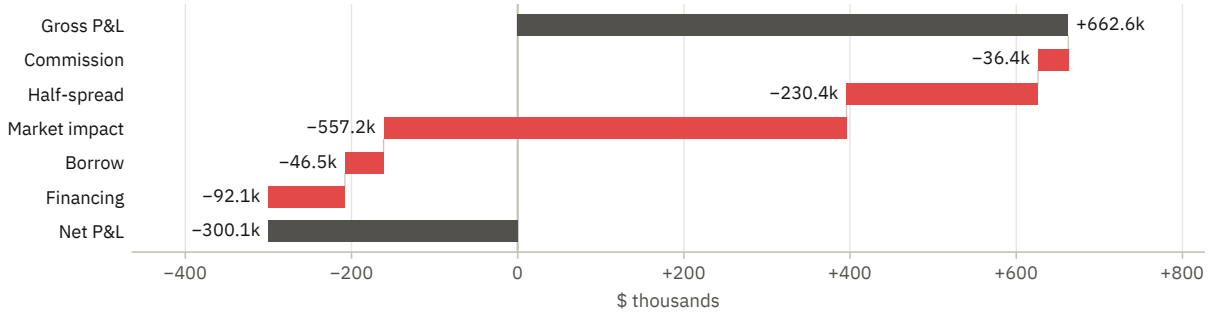


Notes The tick marks the count expected by chance, an upper bound. Pairs share legs, so the tests are dependent; Benjamini–Hochberg is a diagnostic here, not a filter. Selection also requires a half-life of 4–25 days and a quality score. That the 1,775 expected by chance and the 1,776 pairs selected are nearly equal is a coincidence: the second count follows the half-life and quality screens.
Source Author's backtest, v32.0.0 study run (commit 711b18c); prices from Yahoo Finance.

EXHIBIT 12

Gross P&L, itself not significant, does not cover modelled costs; market impact is the largest item

Implementation shortfall of the main backtest: gross P&L to net P&L by cost component, $ thousands



Notes The bars close exactly on the change in the book's value. Gross is 66.3 bp of starting capital, costs 96.3 bp and net −30.0 bp; per unit of traded notional, gross is +8.9 bp and costs 12.9 bp.
Source Author's backtest, v32.0.0 study run (commit 711b18c); prices from Yahoo Finance: costs of each trade by component, and the final portfolio value.

be expected to pass. That count is an upper bound on chance rejections, since it treats every pair as null and the two-direction Bonferroni p-value is conservative, so up to 1,775 of the 2,965, 60%, could be chance. Only 1 pair survives a Benjamini–Hochberg correction at q = 0.05 (Benjamini and Hochberg 1995): the excess is spread thinly over many weak relationships. Selection kept 1,776 pairs, of which 100 were traded (Exhibit 11).

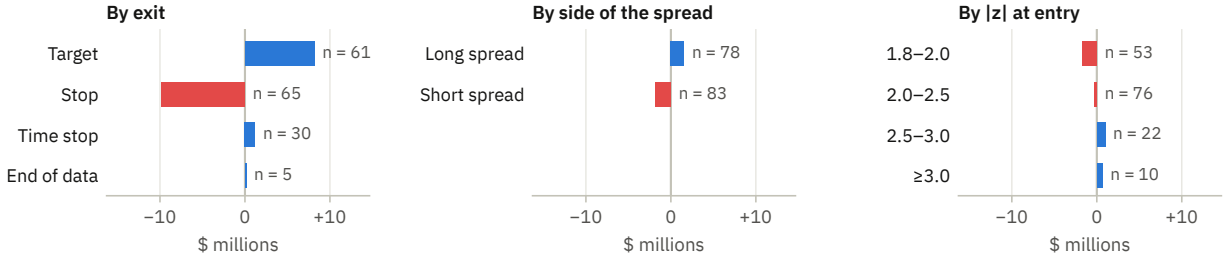
6 Where the edge goes

There is no detectable gross edge for costs to remove. Before costs the 161 trades earned $662.6k, +8.9 bp of the notional they traded (t = 0.28 for the mean gross P&L per trade); modelled costs were $962.7k, 12.9 bp, 57.9% of it market impact (Exhibit 12). Weighting each trade equally gives the same picture: +11.4 bp before costs (t = 0.37) against 12.5 bp of cost. Costs would have to fall to 0.69× their modelled level, the ratio of gross to cost, for net P&L to break even, and the gross return is itself indistinguishable from zero. The trades show the mechanism: the 65 stop-loss exits lost $9.80M, more than the 61 target exits earned ($8.18M; Exhibit 13).

EXHIBIT 13

Stops lost more than targets earned, and short-spread trades lost about what long-spread trades made

Net P&L of the 161 trades by exit reason, side of the spread and |z| at entry, $ millions



Notes Long-spread trades made $1.55M and short-spread trades lost $1.85M, consistent with the net market exposure that hedge ratios below one leave in each trade (Section 8). Win rate 47.2% (76 of 161; 95% interval 39.6–54.9%). The |z| buckets describe the past and are not a selection rule: a higher entry threshold read off this panel would be fitted to the test period.
Source Author's backtest, v32.0.0 study run (commit 711b18c); prices from Yahoo Finance: the 161 trades.

TABLE 6

The v32 cost model against published references, and what v33 changes

Component	v32 model	Reference	v33
Commission	$0.002 per share, $0.50 minimum per order	Author's assumption	unchanged
Half-spread	Half the typical quoted spread by 20-day dollar ADV: 0.75 bp (≥ $500M) to 40 bp (< $5M)	Frazzini et al. (2018)	bucket fit to be checked against measured spreads
Market impact	$Y \cdot \sigma_{daily} \cdot \sqrt{(\text{shares} / \text{ADV})}$, $Y = 0.3$, capped at 50 bp, on entry and exit	The 0.314 of Almgren et al. (2005) is their permanent-impact coefficient, not a trading cost; their temporary-impact fit lies below this model and a square-root law with $Y = 1$ lies above it (the execution-cost note, Section 13)	Y stressed at 0.3, 0.6 and 1.0
Borrow	General-collateral fee, 25 bp a year on the short leg	D'Avolio (2002) on general-collateral fees; the v33 fee is a median buy-side indicative fee, whose source the execution-cost note gives	one 0.38% fee a year replaces borrow plus financing
Financing	Prime-broker spread, 50 bp a year on the short half of gross	Overlaps with the borrow fee (a double count)	folded into the single fee
Fill	Close of $t + 1$ for entry and exit; legs $1 : $β	—	$t + 1$ open as a declared variant

Source Author's cost model for the v32 and v33 study designs; the references cited.

7 Execution and costs

Every fill pays its own costs, from the stock's own liquidity on that day, and fills at the next day's close: a signal seen at the close of day t trades at the close of day $t + 1$. The average trade held 12.6 trading days, and its larger leg was 1.01% of that stock's dollar ADV. Annual turnover was 6.0× capital. Table 6 sets out the cost model against published references and what v33 changes; the study's execution-cost note compares the model with published impact estimates at each participation rate (Section 13).

How much the conclusion depends on the cost model is measured, not asserted. Re-costing and re-sizing the main backtest's own trades under five other cost and sizing schedules (Exhibit 14, Table 7) turns none of them into a significant gain. A cost-sensitivity analysis on daily gross and cost series, with a stress table and a capacity curve, is forthcoming (Table 11).

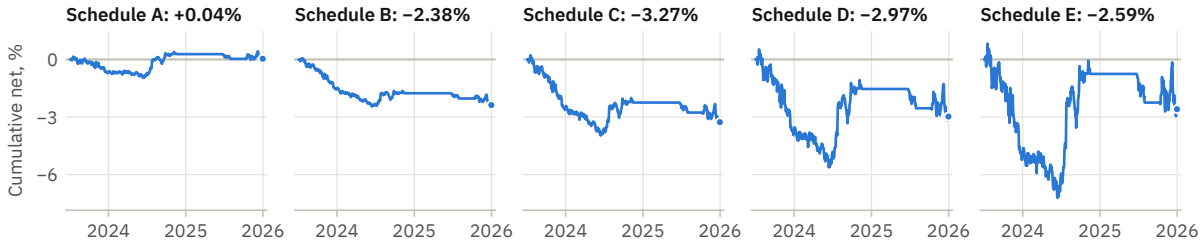
8 Risk and factor attribution

Market exposure is small but not zero. Against an equal-weight sector-ETF proxy the main backtest's beta is 0.010 (t = 1.81) and the walk-forward's 0.016 (t = 2.17; Exhibit 15). The hedge ratio was below one in 148 of 161 trades, and with legs of $1 and $β such a trade is not dollar-neutral: a long-

EXHIBIT 14

Re-costed under five other cost and sizing schedules, the same trades lose money in four and break even in one

Cumulative net return, % of capital, when the main backtest's trades are re-sized and re-costed; schedules A–E in order of total cost



Notes Schedules are parameter sets, not descriptions of any firm; Table 7 lists each one's parameters. Returns run from +0.04% (schedule A, the lowest costs and smallest positions) to −3.27% (schedule C). The replay keeps the main backtest's signals, entry and exit dates and daily paths, and changes sizes and costs.
Source Author's replay of the v32.0.0 trades under schedules A–E.

TABLE 7

Cost schedules A–E: parameters and results of the replay

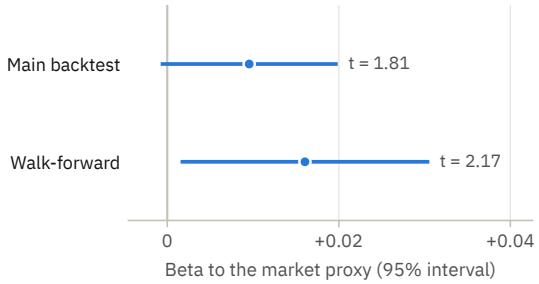
	Comm., bp	Half-spread, bp	Impact cap, bp	Borrow, % a year	Financing, % a year	Position, % of equity	Return, %	Sharpe	Max DD, %	Costs, % of capital	Peak gross, × capital	Trades replayed
A	1.0	1.5	4	0.10	0.00	2–5	+0.04	+0.03	−1.09	0.22	0.15	161
B	0.7	15.0	80	6.00	3.00	2–5	−2.38	−1.58	−2.52	2.57	0.15	152
C	5.0	10.0	50	2.00	1.50	4–10	−3.27	−1.08	−4.17	3.74	0.30	161
D	1.5	4.0	20	1.00	1.00	8–20	−2.97	−0.48	−6.12	3.88	0.60	161
E	1.0	2.0	12	0.60	0.50	12–30	−2.59	−0.27	−7.96	3.88	0.90	161

Source Author's replay of the v32.0.0 trades under schedules A–E; returns, Sharpe ratios and drawdowns as in the run's headline results.

EXHIBIT 15

The only loading measured so far is a small market beta, significant in the walk-forward

Beta to the sector-ETF market proxy with 95% interval (Newey–West)

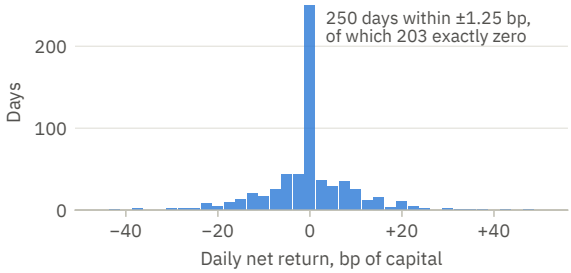


Notes Proxy: equal-weight daily return of 11 sector SPDR ETFs.
Source Author's backtest, v32.0.0 study run (commit 711b18c); prices from Yahoo Finance, including the sector SPDR ETFs.

EXHIBIT 16

Daily returns are fat-tailed (kurtosis 6.7) with a spike at zero from flat days

Distribution of daily net returns, bp of capital; main backtest



Notes Skew 0.17; kurtosis is non-excess (3 for a normal distribution). The book held no position at the close on 207 days; on 4 of them positions were closed that day, so 203 days have exactly zero return. Fat tails widen the Sharpe ratio's standard error, which the power figures in Section 5 account for.
Source Author's backtest, v32.0.0 study run (commit 711b18c); prices from Yahoo Finance.

spread trade is net long and a short-spread trade net short. By notional at the fill, 75 of 78 long-spread trades were net long and 74 of 83 short-spread trades net short; in one trade a hedge ratio just above one was reversed by share rounding. Annualised alpha is −0.25% in the main backtest (t = −0.24) and −0.84% in the walk-forward (t = −0.79). Daily returns are fat-tailed, with a spike at zero from the days the book was flat (Exhibit 16), which widens the Sharpe ratio's standard error. The multi-factor attribution, on the five factors of Fama and French (2015), momentum (Carhart 1997) and short- and long-term reversal with Newey–West errors, is forthcoming (Table 11).

9 A second strategy family: residual reversion

A null on one rule says little about the idea. Avellaneda and Lee (2010) trade each stock's residual against its sector ETF when that residual, modelled as a mean-reverting process, is stretched; many small concurrent bets give the test far more power than 161 pair trades. The parameters are taken from the paper unchanged: a 60-day regression on the sector ETF, an AR(1) fit of the cumulative residual, trades only when mean reversion is faster than 30 days, entry at $|s| > 1.25$, exits at −0.50 (long) and 0.75 (short), next-day fills and the same cost model. The arm runs on the development calendar with the v33 code and then, once, on the holdout (Section 10). It has not yet run on the development calendar; its result is forthcoming (Table 11).

10 The 2026 holdout test

Every design choice in this study was made on 2020–2025 data over 32 versions, so the development period cannot test those choices. The decisive test therefore runs on data none of them has seen: trading days from 2 January 2026. No 2026 daily price series has been fetched for any stock or ETF. Two inputs do carry 2026 market information: the v32 universe ranks stocks by market capitalisation on 18 October 2026, and the factor files of the Kenneth French library extend into 2026. No strategy reads factor data, and every run truncates the files at its own calendar's last day; the ranking sets the frozen v32 arm's membership in 2026, a look-ahead disclosed in Table 10. The protocol is committed first; each arm then runs once, and its result is published whatever it is. At the date of this note the pre-registration is not yet filed; Table 8 gives the hypotheses it is to fix.

TABLE 8

Hypotheses proposed for the pre-registration, to be filed before any 2026 price series is fetched

	Arm	Hypothesis
H1	v33 pairs (design proposed for pre-registration)	mean daily net return > 0
H2	Avellaneda–Lee residual reversion	mean daily net return > 0
S3	v32 as published, frozen	reported, not tested

Notes Test: one-sided Newey–West t on daily net returns, Holm-adjusted across H1 and H2 at 5%. The frozen v32 arm (S3) is reported with the same statistics but tested in no family: its universe is ranked on market values inside the test period, so a rejection could reflect survivorship rather than an edge.

For each arm k the null hypothesis is that the mean daily net return over the holdout is at most zero, against the alternative that it is positive. The test is the one-sided Newey–West t-test on the daily net return series; the two confirmatory p-values (H1 and H2) are adjusted with the step-down procedure of Holm (1979) at a familywise level of 5%, and an arm shows an edge out of sample only if its adjusted p-value is below 0.05. The frozen v32 arm reports the same statistics but enters no test, because its universe is ranked inside the test period. Whatever the outcome, each arm reports its net and gross return, its Sharpe ratio with standard error, the probabilistic Sharpe ratio, the block-bootstrap interval, the one-sided upper bound, trades and exposure, and, for the pairs arms, the Ledoit–Wolf difference against the Gatev method and the random-pair percentile on the same holdout.

Power, stated in advance. About ten months of data can detect only a large Sharpe ratio. With returns shaped like the development period's (skewness

TABLE 9

Smallest annual Sharpe ratio the holdout detects with 80% power at Holm's first step (one-sided 2.5%)

Holdout length, trading days	Returns shaped like the development period's	Independent normal returns
126	4.3	4.0
200	3.4	3.2
252	3.0	2.8

Notes Six months, the registered calendar of 2 January to 19 October 2026, and a year. The development shape uses the asymptotic Sharpe variance for non-normal returns with lag-1 autocorrelation, as in Table 4; it is computed from development returns only.
Source Author's calculation from the daily net returns of the v32.0.0 run.

0.17, non-excess kurtosis 6.7, lag-1 autocorrelation 0.07), a test at Holm's first step (one-sided 2.5%, the level the better of the two arms must reach) with 80% power detects an annualised Sharpe ratio of about 3.4 over the registered 200 trading days and 3.0 in a full year (Table 9). The holdout is therefore a falsification check, not a proof of an edge: it can reject a large edge but cannot confirm a small one, and not rejecting the null is not evidence that the edge is zero, which is why each arm's upper confidence bound is reported. Running v32 alone first would spend the holdout, so v33 is to be frozen before any 2026 price series is fetched, and all three arms run once; two are tested.

11 Threats to validity

Each limitation, which way it biases the result, how large it is, and what addresses it (Table 10). Biases that flatter the strategy make the null more convincing, not less.

TABLE 10

Threats to validity of the v32 study run and the 2026 test

Threat	Direction of bias	Size	Mitigation
Survivorship, and an approximate index	↑ flatters	Unmeasured: the universe is the 3,000 largest stocks of 18 October 2026, back-filled, so pairs broken by failures and delistings are missing	v33: point-in-time membership from IWV holdings in SEC N-PORT filings, with the coverage gap at each formation reported (forthcoming, Table 11)
Design choices were made on the same data	↑ flatters	32 versions on 2020–2025; deflated Sharpe with 32 trials is a lower bound	2026 holdout, to be pre-registered; declared variant family with SPA, stepdown, MCS and PBO
The walk-forward reuses tuned thresholds	↑ flatters	Entry, exit and stop thresholds and the regime gate were tuned on data inside the walk-forward	Called walk-forward, not out of sample; the holdout is the out-of-sample test
Re-selection can drop pairs but not add them	↓ penalises	The pool can only shrink after 2022	v33: full re-formation every 126 trading days on the trailing 3 years
Borrow is general-collateral for every name	↑ flatters	Borrow was $46.5k of $962.7k costs; specials and recalls are not modelled	v33: 0.38% buy-side median fee; hard-to-borrow stress
Fills are close-to-close	· either way	Larger leg 1.01% of ADV on average; no auction or queue model	$t + 1$ open variant; closing-auction cost model planned
Ranker features include pair statistics estimated on the formation window	↑ flatters	Small: out-of-sample AUC 0.514, no useful skill to inflate	v33: the ranker becomes an ablation arm only
The neural ranker sees one 30-feature vector	· none on P&L	A sequence of length one: functionally a residual multilayer perceptron	v33: L2 logistic and gradient-boosted trees
Sector features are coarse	↓ penalises	Pairs are not restricted to one industry, so many are economically unrelated	v33: pairs within one Fama–French 49 industry
The 126-day re-selection check has little power	↓ penalises	In simulation it passes about a quarter of non-cointegrated pairs and drops a third to two thirds of cointegrated ones	v33: replaced by full re-formation
The price cache used through v31 was built with a full-sample penny-stock screen	· none for v32	The v32 run used a freshly fetched v32 cache	Fetch manifest and coverage check in every run
"Point-in-time" prices undo splits, not dividends	· small, either way	Early prices are understated by later dividends in the $5 price and dollar-volume screens	v33 keeps the raw close
Splits after the data window cannot be undone	· small, either way	The vendor back-adjusts history for every split up to the fetch date but reports split events only inside the requested window,	Measured once the holdout opens and those split events can be fetched; disclosed

Threat	Direction of bias	Size	Mitigation
		so names that split after it are mis-screened; returns, z-scores and hedge ratios are unaffected (a constant factor)	
The regime gate was tuned on windows now inside the test period	· either way	New entries blocked on 248 of 628 days	v33: off by default; counted as a declared variant
Hedge ratios below one leave market exposure	· adds beta	Beta t = 2.17 in the walk-forward (Exhibit 15)	v33: multi-factor attribution
The main backtest and the walk-forward overlap	↑ flatters	616 of 628 days in common, same pair pool	The holdout is the independent test
The random-pair control trades a different rule	· weakens the comparison	Random pairs run through the Gatev rule, not this engine	A control on the strategy's own engine is planned
The v32 arm's universe in the holdout is membership as of 18 October 2026	↑ flatters	Look-ahead in membership for the frozen v32 arm (H3) during 2026	To be disclosed in the pre-registration; the v33 and Avellaneda–Lee arms use point-in-time membership
Factor files from the Kenneth French library extend into 2026	· none on trading	No strategy reads factor data	Every run truncates them at its own calendar's last date; no development export holds a 2026 factor return
Renamed securities with a new CUSIP look like delistings	· overstates the gap	Counted in the measured survivorship gap, which is therefore conservative	The data review remaps through N-PORT tickers and issuer successions

Notes ↑ flatters the strategy; ↓ penalises it; · no single direction.

12 Forthcoming results

Each analysis below is specified, and its exhibit or table is built, but its results are not yet among the committed data this note reads. When they are, the item leaves this table and appears in its section. None of them is reported here as a number.

TABLE 11
Forthcoming results

Analysis	What it will show	Awaits
Known-answer validation	The engine and its full cost model on synthetic markets with a known answer: the gross edge on planted cointegrated pairs, no edge on forced non-cointegrated pairs, the hedge-ratio error and pair-selection recall	A committed validation run
Point-in-time universe coverage	Index members with and without price history at each formation date, and the index weight the price data do not cover: the survivorship gap	The v33 development run
Engle–Granger p-values	The distribution of the 35,491 p-values against the uniform distribution expected if no pair were cointegrated	The v33 development run
Data-snooping tests	White's Reality Check, Hansen's SPA test, the Romano–Wolf stepdown, the model confidence set and the probability of backtest overfitting over the declared family (27-point grid, 11 toggles)	The family's run on the development calendar
Cost sensitivity	Net Sharpe ratio from 0× to 2× modelled costs with the break-even multiple, the Sharpe ratio gross of costs, and the impact coefficient at 0.3, 0.6 and 1.0 crossed with two borrow fees	The v33 development run
Exit race	Targets against stops, compared with the count expected for a spread that does not mean-revert	The v33 development run
Decay from formation to trading	Gross return per trade of the same rule on the same pairs, formation period against trading period	The v33 development run
Persistence of cointegration	The share of selected pairs still cointegrated in later windows, against the 5% expected by chance	The v33 development run
Capacity	Impact cost against book size for a stated hypothetical gross edge (the measured edge is not significant)	The v33 development run
Multi-factor attribution	Loadings on the five Fama–French factors, momentum and short- and long-term reversal, with Newey–West errors	The v33 development run
Random-pair band	Each day's 5th–95th percentile of the random-pair draws' cumulative returns, drawn around the equity curve	The v33 development run
Avellaneda–Lee residual reversion	The second strategy family's development-period net Sharpe ratio and 95% interval, beside the pairs strategy	Its run on the development calendar
The 2026 holdout	Each arm's net Sharpe ratio, Holm-adjusted p-value and decision (H1 and H2; the v32 arm descriptive), and its cumulative return	The pre-registration, then a single run on 2026 data

13 Reproducibility

The research-note site presents the same results; Table 19 maps its exhibit numbers to this note's. Every number in this note is formatted by

`scripts/build_paper.py` from `assets/study.json`, the file the research-note site's charts and checked numbers come from, and from the headline JSON that file was built from; the site's numbers are checked against the same JSON by `scripts/check_published_numbers.py`. Prices come from Yahoo Finance's chart API, and a later fetch will not match the cached data bit for bit, because vendors revise history. The run manifest (Table 17) pins what was used, and the sha256 of every committed source the note reads is listed in Table 18; both are in the appendix.

The run's universe file is not redistributed. `scripts/build_universe.py` rebuilds a universe from the Nasdaq screener on the day it runs, which gives that day's membership, not the list of 18 October 2026; the commands below instead recreate the run's 2,999 symbols from its fetch manifest. The execution-cost note, `docs/EXECUTION_COSTS.md`, compares the cost model with published impact estimates at each participation rate and gives the source of the v33 borrow fee; `tests/test_cost_model_zoo.py` keeps the model inside that band. The AR(1) variance of the Sharpe ratio and its source are documented in `pairs_trading/power.py`, and the synthetic validation is `scripts/synthetic_validation.py`.

```
# the study: the v32 design on the development calendar
pip install -r requirements.txt
mkdir -p data
python - <<'EOF'      # the run's universe, from its fetch manifest
import json
symbols = sorted(json.load(open('results/v32_dev/price_fetch_manifest.json'))['outcomes'])
open('data/marketcap.csv', 'w').write('Symbol\n' + '\n'.join(symbols) + '\n')
EOF
python -m pairs_trading.main --profile v32 --calendar dev   # writes outputs/

# publish a run: copy outputs/ to results/<run>/, then
python scripts/build_site_data.py --dev results/v32_dev
python scripts/check_published_numbers.py --json results/v32_dev/headline_metrics.json

# this paper (needs the optional typst package: pip install -e ".[paper]")
python scripts/build_paper.py           # data, figures and PDF
python scripts/build_paper.py --check   # exit 1 if any of them is stale
```

Disclosures

Hypothetical, backtested results on $100M of notional capital, not achieved by any account. Returns are net of the modelled costs described in Section 7 unless labelled gross, earn no interest on cash, and are shown beside baselines run under the same costs. Hypothetical results have inherent limitations, including those listed in Section 11. This is a research project: not investment advice, not an offer of any product, and no claim of deployable performance. The cost schedules are parameter sets, not descriptions of any firm. Code under the MIT licence; IBM Plex typefaces under the SIL Open Font License 1.1.

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Appendix

TABLE 12

Specification of the v32.0.0 study run

Stage	Rule
Universe	The 3,000 largest US common stocks by market cap on 18 October 2026 (the Russell 3000 rule applied to the Nasdaq screener); 2,752 of its 2,999 symbols have prices; daily prices 2020–2025; screens on point-in-time (split-unadjusted) price and dollar volume; the 300 most liquid enter the pair search.
Selection	Positive return correlation ≥ 0.25; two-direction Engle–Granger $p < 0.05$ (Bonferroni across directions); half-life 4–25 days on the cointegrating spread; quality score. Every 90 trading days a re-test on the trailing 2 years, with the Engle–Granger p-value on the last 126 calendar days at most 0.30; pairs can drop out, not enter.
Spread	Two-state Kalman filter (slope and intercept) on log prices centred on their burn-in means, so β is a hedge ratio and signals are invariant to vendor back-adjustment.
Rule	One β-locked z-score against its mean and standard deviation over the last 30–60 bars, frozen at the signal. Entry $\|z\| \geq 1.8$; target once z reverts past −0.5 (long) or +0.5 (short); stop 1.5 further against; time stop clamp(2.5 × half-life, 10, 25) days. Gates: 20-day correlation ≥ 0.20, β in [0.25, 4], a 30-day bar after three consecutive losses, VIX and sector-dispersion regime scaling.
Book	$100M notional; 3–10% of equity per pair before regime scaling (1.2% to 9.0% as traded); 30% gross cap; at most 8 entries a day; no shared legs; daily mark-to-market; drawdown and daily-loss kill switch.
Ranker	A classifier of P(trade profitable) on 30 features, used only to order same-day candidates; out-of-sample AUC 0.514 (0.510 pooled across walk-forward windows; 0.5 is no skill).
Walk-forward	One continuous simulation on a 252-day train / 63-day test grid, the last window running to 12 December 2025; the ranker is refit per window; per-window figures are calendar slices.
Evidence	Probabilistic and deflated Sharpe, Newey–West t, Politis–White block bootstrap, Lo (2002) autocorrelation-adjusted annualisation, Ledoit–Wolf Sharpe-difference test, PBO; Gatev distance method and random pairs under the same costs and $t + 1$ fills.

TABLE 13

The v32 design and the v33 design proposed for pre-registration

Choice	v32 (run reported here)	v33 (proposed)
Universe	the 3,000 largest US stocks by market cap on 18 October 2026	point-in-time members (IWV N-PORT)
Pair search	300 most liquid names, all pairs	500 most liquid names, same Fama–French 49 industry
Re-selection	re-test the 2022 pool every 90 trading days	full re-formation every 126 trading days on the trailing 3 years
Regime gate	on	off (a declared variant)
Ranker	learned classifier orders same-day candidates	none (classical ranking; ablation arm)
Borrow and financing	25 bp borrow + 50 bp financing a year	one 0.38% fee a year

Notes None of the v33 choices was made by its effect on development-period P&L; the ranker was dropped for its lack of predictive skill on that period. The reasons are stated next to each setting of the study's design profiles.

TABLE 14

Variants declared on 20 October 2026 for the data-snooping tests

Variant	What it changes
`ENTRY_Z`	grid values 1.50, 1.80, 2.10
`EXIT_Z`	grid values 0.25, 0.50, 0.75
`STOP_Z`	grid values 1.00, 1.50, 2.00
`regime_gate_on`	the v10/v11 gate fitted on now-test windows
`cusum_gate_on`	entry gate tried in earlier versions
`hurst_gate_on`	entry gate tried in earlier versions
`no_rolling_corr_floor`	decoupling filter (20-day correlation ≥ 0.20) removed
`ref_bars_40`	z-score reference window, shorter
`ref_bars_90`	z-score reference window, longer
`ranker_gbm`	the learned ranker (gradient-boosted trees) switched on
`open_fills`	t + 1 open instead of t + 1 close fills
`all_pairs`	no same-industry restriction
`universe_300`	v32's 300 most liquid names
`reform_every_63`	quarterly instead of half-yearly re-formation

Notes The threshold grid covers the three hand-set trading thresholds, which were chosen across versions on this same data. Each toggle changes one setting of the v33 design.
Source The declaration of the variant family, committed on 20 October 2026.

TABLE 15

Walk-forward windows: calendar slices of one continuous simulation

Window	Training from	Test from	Test to	Days	Net return, %	Sharpe	Trades opened	Ranker AUC
1	11 Feb 2022	14 Feb 2023	15 May 2023	63	−1.68	−2.50	42	0.511
2	13 May 2022	16 May 2023	15 Aug 2023	63	+1.30	+2.39	41	0.506
3	15 Aug 2022	16 Aug 2023	13 Nov 2023	63	−0.14	−0.27	34	0.557
4	11 Nov 2022	14 Nov 2023	14 Feb 2024	63	−0.23	−0.44	22	0.565
5	14 Feb 2023	15 Feb 2024	15 May 2024	63	+0.09	+0.20	23	0.518
6	16 May 2023	16 May 2024	15 Aug 2024	63	−0.31	−0.46	31	0.517
7	16 Aug 2023	16 Aug 2024	13 Nov 2024	63	−0.15	−0.33	13	0.503
8	14 Nov 2023	14 Nov 2024	18 Feb 2025	63	−0.12	−2.85	0	0.525
9	15 Feb 2024	19 Feb 2025	19 May 2025	63	0.00	—	0	0.560
10	16 May 2024	20 May 2025	19 Aug 2025	63	−0.73	−2.80	1	0.487
11	16 Aug 2024	20 Aug 2025	12 Dec 2025	81	+0.24	+0.50	12	0.497

Notes Windows are numbered in calendar order. Each window's ranker is trained on the year before its test start. Positions carry across window boundaries, so a window that opens no trade can still carry P&L from earlier positions (window 8). Sharpe ratios are annualised and not defined for a window whose daily returns are all zero (window 9). The last window runs to the walk-forward's last day.
Source Author's backtest, v32.0.0 study run (commit 711b18c); prices from Yahoo Finance.

TABLE 16

Selected versions of the study

Version	Change
v9–v12	First look-ahead fixes (P&L, pair selection, Kalman β drift).
v16–v19	Kalman spread fixes, β-weighted P&L, PCA residual cointegration, dynamic hold.
v24–v26.1	Ranker trained and wired in (ranking only); controlled ablation; a label bug that had disabled training.
v28–v29	Significance layer and baselines; $t + 1$ entry fills.
v30	Leakage audit: survivorship confirmed, selection-clean window tagging.
v31–v31.1	Pre-launch audit: inverted stop-loss, same-bar exits, future re-selections in the walk-forward; results withdrawn.
v32	Full audit and rebuild of the engine, Kalman model, accounting, walk-forward, costs, ranker, benchmarks and outputs; synthetic known-answer validation; study re-run: no detectable edge.
v33 (in progress)	Design proposed for pre-registration: point-in-time universe, same-industry pairs, full re-formation, regime gate off, one borrow fee, a second strategy family, the evidence layer and the sealed 2026 holdout.

Notes Versions that only withdrew or corrected earlier results are listed with the fixes they made.

TABLE 17

Run manifest

Item	Value
Commit	`711b18cbd1a6 (clean)`
Run	`2026-10-18 09:21 UTC`
Price cache	`sha256 70776501502e08533cf17d46bbf466b26bb9d68233aa467aa1665d4ea5525ea1`
Headline JSON	`results/v32_dev/headline_metrics.json, sha256 934715640095540 07e8bff1f6a67a02f7f2db96b99d76ad8bd3e5a57b5183b0e`
Site data	`assets/study.json, sha256 0e29723453ec72cbc84e0b52709ff1cc7fc87529c63a20a592382208e790f676`
Universe	`Nasdaq screener, 18 October 2026; 2,752 of 2,999 symbols with prices`
Seed	`PAIRS_SEED=42`
Stack	`python 3.12.3 · numpy 1.26.4 · pandas 2.2.3 · scipy 1.15.2 · statsmodels 0.14.4 · torch 2.13.0+cpu`

TABLE 18

Committed sources read by this note

File	sha256
`results/v32_dev/fund_type_comparison.json`	`1ad3925715acbe90861dc5d3b8561266dc38e36b8e87a43f20e93f6339a649a4`
`results/v32_dev/headline_metrics.json`	`934715640095540 07e8bff1f6a67a02f7f2db96b99d76ad8bd3e5a57b5183b0e`
`results/v32_dev/pairs_trading_results_2023_2025_unbiased.json`	`0d2e376fdbac5de777aa33187938d9e0d23ef76207f4cdc92de93ce260f9e876`
`results/v32_dev/universe_meta.json`	`af2bc3d41d0b2cee0381c6479baa144d1a743cde7f78dc266de3bddac20116b5`
`results/variant_family/declaration.json`	`028b1a29935565931fbe191ac578d8f2756d36b1ddc1d7ecb4833b8f7ca66add`
`results/v32_dev/price_fetch_manifest.json`	`5f6b142038e794ebf767243c9f3f0ffd5111e5915141039fc5564ab1e85a00b5`
`reference/universe/index.json`	`d5bfbe0fdc4e0b52c15ac0767886fa40983010834e7959d81cb06647e0552f06`
`index.html`	`887ef32af50a44688c288955c73f05cbae85cea9ee3a5bbc408e75e495dadfb3`

Notes The PDF is compiled with a fixed timestamp, the note's date, so a rebuild of unchanged inputs is byte-identical; the build manifest records the sha256 of every compile input and of the PDF.

TABLE 19

Exhibits of the research-note site and where they appear in this note

Site	Title on the site	This note
Exhibit 1	Net Sharpe ratio of each pre-registered strategy on the 2026 holdout	forthcoming (Table 11)
Exhibit 2	Net of costs the book ends slightly below where it started; the Gatev baseline ends at +0.57%	Exhibit 4
Exhibit 3	Out of sample, 3 of 11 walk-forward windows made money; the stitched path ends at −1.75%	Exhibit 5
Exhibit 4	One month carries the result: July 2024 earned +2.14%; the whole period returned −0.30%	Exhibit 6
Exhibit 5	Half-year Sharpe ratios swing between −2.8 and +3.4; the only stretch whose band clears zero contains July 2024	Exhibit 7
Exhibit 6	Every Sharpe interval spans zero; the data exclude only Sharpe ratios above about 1	Exhibit 8
Exhibit 7	The strategy sits at the 28th percentile of 200 random-pair portfolios	Exhibit 9
Exhibit 8	Deflated for 32 trials, the main backtest's Sharpe ratio has a 1.4% probability of beating chance; the data-snooping tests are pending	Exhibit 10
Exhibit 9	Gross P&L, itself not significant, does not cover modelled costs; market impact is the largest item	Exhibit 12
Exhibit 10	Stops lost more than targets earned, and short-spread trades lost about what long-spread trades made	Exhibit 13
Exhibit 11	Sharpe ratio against a multiple of modelled costs	forthcoming (Table 11)
Exhibit 12	Exit race: targets against stops, and against a spread with no mean reversion	forthcoming (Table 11)
Exhibit 13	In-sample to out-of-sample decay of the same rule on the same pairs	forthcoming (Table 11)
Exhibit 14	Do selected pairs stay cointegrated?	forthcoming (Table 11)
Exhibit 15	Re-costed under five other cost and sizing schedules, the same trades lose money in four and break even in one	Exhibit 14
Exhibit 16	Capacity: impact cost against book size for a stated gross edge	forthcoming (Table 11)
Exhibit 17	The only loading measured so far is a small market beta, significant out of sample	Exhibit 15
Exhibit 18	Daily returns are fat-tailed (kurtosis 6.7) with a spike at zero from flat days	Exhibit 16
Exhibit 19	From 35,491 candidate pairs to 100 traded; chance alone accounts for most “cointegrated” pairs	Exhibit 11
Exhibit 20	Engle–Granger p-values against the uniform null	forthcoming (Table 11)
Exhibit 21	Avellaneda–Lee residual reversion against the pairs strategy, development period	forthcoming (Table 11)
Exhibit 22	The pipeline, and what data each stage may see	Exhibit 2
Exhibit 23	Data splits: formation, walk-forward, main backtest and the sealed holdout	Exhibit 3
Exhibit 24	Point-in-time universe: index members with and without price history at each formation date	forthcoming (Table 11)
Exhibit 25	Gross bp per trade on synthetic markets with known answers	forthcoming (Table 11)

Notes The site orders its exhibits by its own sections. Titles are the site’s; where this note words a finding differently (for instance, it calls the walk-forward out of sample only for pair selection and the ranker), this note’s wording applies. Exhibits of this note without a site counterpart are not listed.